

Project 2G

Singularity Analysis of a New Remote Center of Motion Pivot Mechanism

Motivation and Context:

The KITE pivot is a new kinematic architecture for a remote center of motion (RCM) mechanism. Its planar topology is based on a serial arrangement of a revolute joint and a parallelogram linkage. This module is repeated radially and connected to the moving stage, creating two parallel modules that together constrain two DoF and leave a single DoF for the moving stage.

For a given topology (see Fig. 1), this single DoF can be a pure rotation about a remote center. The only condition is that the "virtual" quadrilaterals $OBCD$ share a vertex and are parallelograms. This gives rise to a multitude of possible configurations, including overconstrained ones (see Fig. 1 and 2).

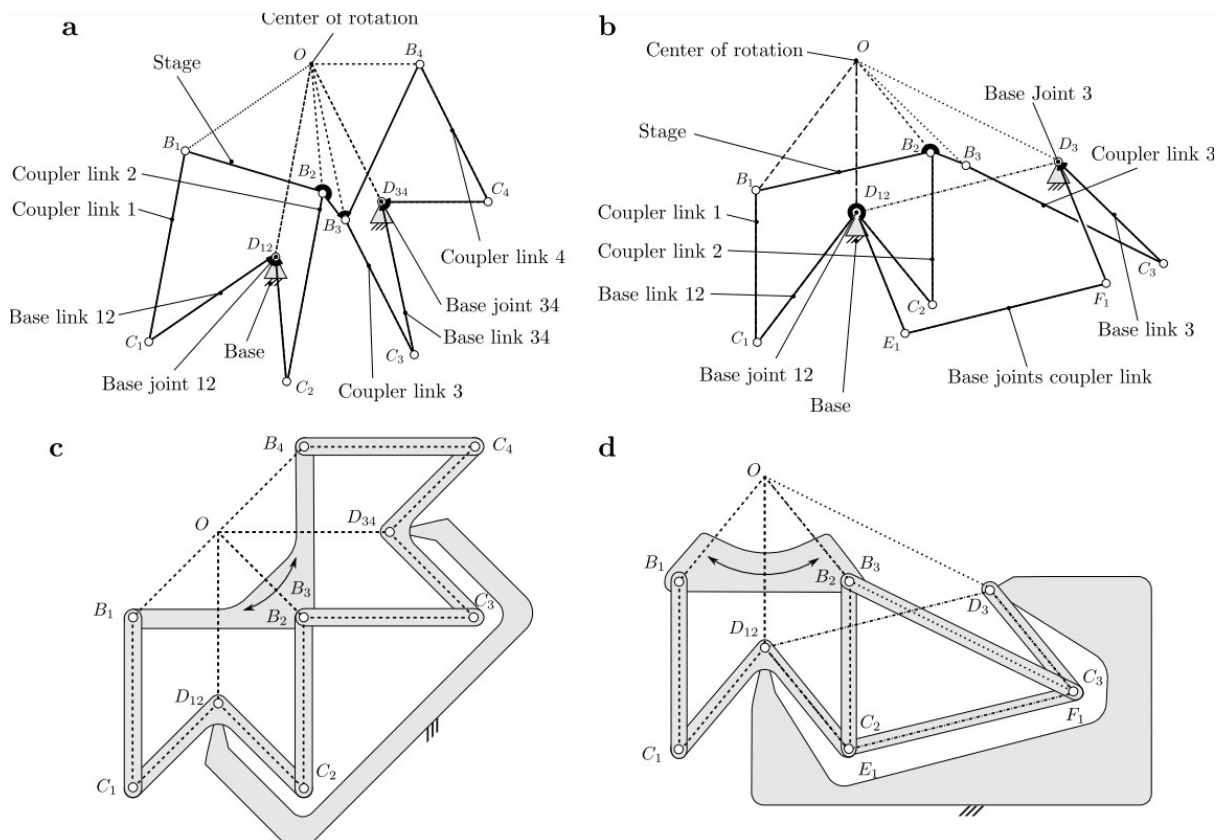


Figure 1: Generic designs of geometrically exact RCM pivot linkages (a) Four coupler links configuration. (b) Three coupler links and one homokinetic coupling configuration. (c) (d) respective rigid-body implementations options with coincident joints.

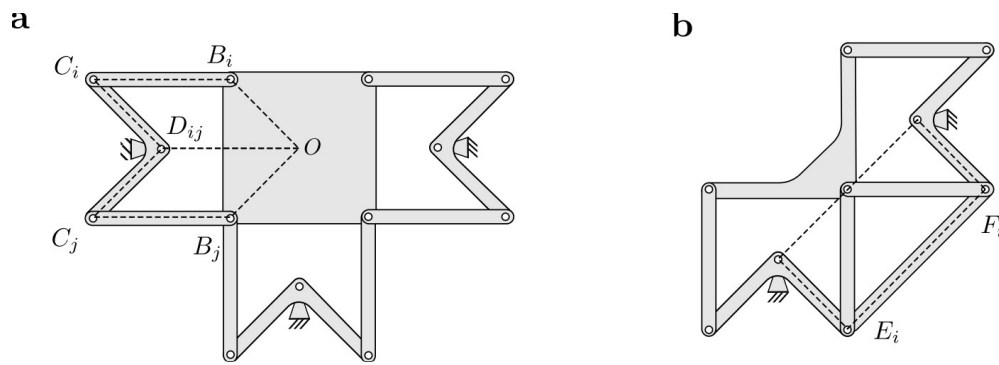


Figure 2: Design of two overconstrained single DoF mechanisms with one redundant constraint $R=1$. (a) Additional base revolute joints and associated coupler links. (b) Additional homokinetic coupling between to base li

The KITE pivot was initially developed as a flexure-based RCM mechanism with minimized parasitic shift (see Fig. 3), whose angular stroke is inherently limited. However, a rigorous analysis of the underlying rigid-body linkage is still lacking, especially for applications requiring unlimited range of motion. This gap motivates the present work.

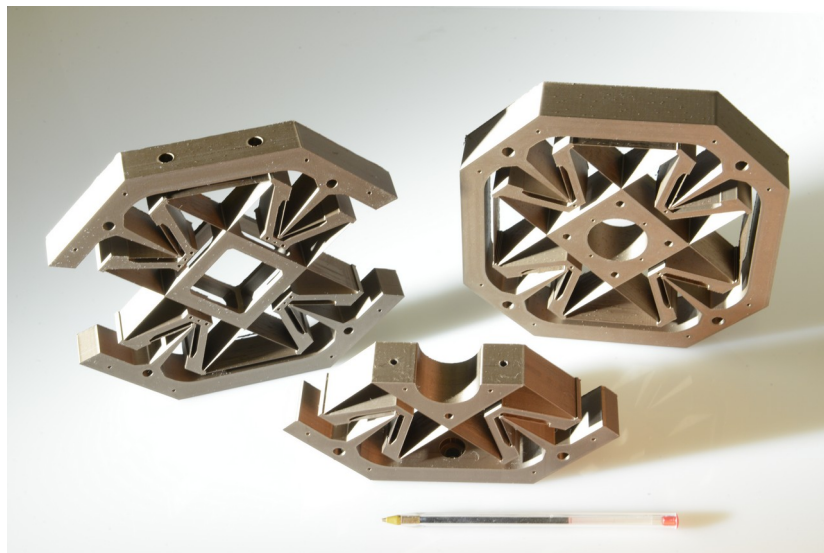


Figure 3: The flexure-based KITE pivot family

Tasks

Overall objectives. Perform a singularity analysis of the entire robot. Try to show that the moving stage can undergo unlimited rotation with appropriate strategies.

I. Direct, inverse kinematic

Compute analytical forward and inverse kinematics. Then prove the existence of a perfect rotation of the end defector without parasitic motion if all quadrilateral OBCD are parallelograms. Bonus: prove the uniqueness.

Simscape model

Use this model in order to verify the kinematics equations. Geometrical parameters must be chosen such that unlimited rotation is feasible.

II. Singularity analysis:

This mechanism is prone to several types of singularities, whose nature depends on the initial geometry. To identify them, the Jacobian matrices of the KITE kinematics should be computed for different configurations. The first task is to list the types (I or II) that can appear in the mechanism. Owing to its overconstraint and initial topology, these singularities are expected to be avoidable through "out-of-phase" parallelogram couplings, additional coupling links, or both. Avoidance strategies are then proposed.

III. Unlimited rotation design:

The objective is to design an RCM mechanism capable of a full 360° rotation while avoiding both singularities and link collisions.

IV. Joint Torque. Consider quasi-static assumption with mass associated to the end-effector and gravity.

Expected outcome Students should obtain the forward and inverse kinematics and the Jacobian matrices of the KITE kinematic. They should develop a singularity avoidance strategy that enables unlimited rotation. Submit a report, and a short video (screen recording) with Simscape illustrating unlimited rotation.

Additional Content

Visualization of the singular configuration of the KITE from Fig.1c

